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*To my parents
and friends*



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Abstract

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Introduction

1.1 Motivation

The number of wireless sensor networks (WSNs) and Internet of Things (IoT) devices has grown rapidly in recent years, raising a fundamental question: how to power all these devices without replacing batteries or requiring constant maintenance. Recent advances in low-power electronics have reduced the energy consumption of sensor nodes to the range of tens to hundreds of microwatts [7]. At such low power levels, it becomes possible to power these devices by extracting energy from the surrounding environment, removing the need for batteries and allowing the network to operate indefinitely.

Batteries are the most common power source for WSN nodes, but they have well-known limitations. In many applications, such as structural health monitoring, industrial machinery surveillance, or medical implants, replacing or recharging depleted batteries is difficult or simply not possible [4]. Using larger batteries to extend the device lifetime increases the size of the system, which is often not acceptable. In addition, the environmental cost of battery and the consequent maintenance of large-scale networks are serious obstacles to the widespread use of WSNs [5].

Energy harvesting offers a practical solution to these problems. Instead of storing a fixed amount of energy, harvesting systems collect energy from sources available in the environment such as sunlight, heat gradients, mechanical vibrations, or radio frequency signals and convert it into electrical power to supply an electronic system.

Among all available energy sources, mechanical vibrations have received the most attention in the research community, accounting for more than half of

publications in this field [5]. Piezoelectric, triboelectric, and electromagnetic transducers are the most widely used mechanisms, each with different trade-offs in output voltage, power density, and complexity [7, 4]. Figure 1.1 shows the distribution of recent publications about transducer types, highlighting the dominance of mechanical energy harvesting approaches.

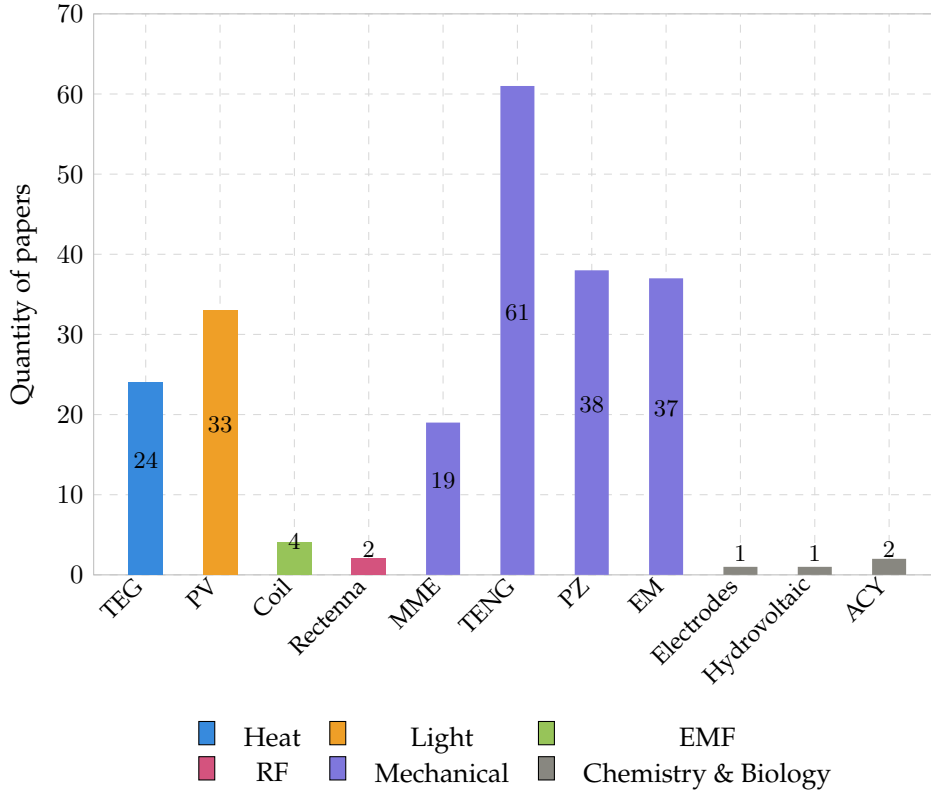


Figure 1.1: Distribution of papers based on transducer type [5].

Energy harvesting systems are generally divided into two categories [4]: systems that use the harvested energy immediately (*Harvest-Use*), and systems that store it first in a battery or supercapacitor before supplying it to the node (*Harvest-Store-Use*). The second approach is far more common, since most energy sources are not continuously available and some form of storage is needed to guarantee uninterrupted operation.

1.2 Energy Harvesting Systems

An energy harvesting system is made of three components that work in sequence, as illustrated in Figure 1.2.

- *Transducer*: converts the ambient energy into an electrical signal.
- *Power conditioning circuit*: perform AC-DC conversion, impedance matching and voltage regulation. A typical implementation combines a rectifier circuit with a power management integrated circuit (PMIC) [9].
- *energy storage element*: is used to keep the device operating for a certain amount of time even if the source is weak or no more present. Such element can be a capacitor/super capacitor or a rechargeable battery [7].

Figure 1.2: Block diagram of a typical energy harvesting system.

The performance of the entire system depends in particular on the transducer. Its design must be matched to the characteristics of the ambient source as well as to the power need of the target device. A transducer that is well suited to one environment may perform poorly in another, which is why the choice of transduction mechanism is the central design decision in any harvesting application [4].

Designing a good harvesting system is not just about getting the most power out of it. The system must work reliably in real conditions, where the energy source can be irregular or unpredictable. At the same time, it must be small, cheap, and easy to integrate into the target device [4]. Finding the right balance between these requirements is the main challenge in this field, and it has led to the development of many different solutions, as discussed in the following sections.

1.3 Performance metrics

A key challenge in evaluating energy harvesting systems is choosing the right performance metrics. Xu et al.[9] propose two metrics for this purpose. The first is the *power conversion efficiency* (PCE), which measures how much of the power entering the interface circuit actually reaches the output:

$$\text{PCE} = \eta_{\text{Rec}} \cdot \eta_{\text{DC-DC}} = \frac{P_{\text{out}}}{P_{\text{in}}}. \quad (1.1)$$

The second is the *power extraction efficiency* (PEE), which compares the output power to the maximum power the transducer could deliver under ideal conditions:

$$\text{PEE} = \frac{P_{\text{out}}}{P_{\text{max}}} = \frac{P_{\text{in}}}{P_{\text{max}}} \cdot \text{PCE}. \quad (1.2)$$

The difference between the two is important. A circuit can convert power efficiently (high PCE) while still wasting much of what the transducer could produce, if the circuit itself is not able to work in the optimal operating point. For this reason, PEE is a more useful metric when the goal is to maximise the total power delivered to the load [9].

1.4 Main Transduction Technologies

Mechanical vibrations can be converted into electrical energy through four main transduction mechanisms: piezoelectric, electromagnetic, electrostatic, and triboelectric. Each of them works in a different way and comes with its own advantages and limitations [6, 2].

Piezoelectric

Piezoelectric transducers produce a voltage when a material is bent or squeezed. The most common design is a cantilever beam with a piezoelectric layer on top: when the beam vibrates, the material deforms and generates an electric charge. This approach gives a high output voltage and does not need a magnet or moving parts, which makes it easy to miniaturise[1]. The main drawback is that piezoelectric materials are brittle and may crack under large deformations. Also, the output drops sharply when the vibration frequency shifts away from the resonant frequency of the beam [7].

Electromagnetic

Electromagnetic transducers generate a voltage by moving a coil near a permanent magnet. This is the same principle used in large electrical generators, applied at a small scale. These harvesters tend to produce a low voltage but can deliver a relatively high current, which makes them a good choice when power output is the main goal. One limitation is that the coil and magnet must move relative to each other, which can be hard to achieve in applications involving large or slowly rotating objects [6].

Electrostatic

Electrostatic transducers work by changing the capacitance between two conductive plates as they move. When the plates get closer or further apart due to vibration, the stored electrical energy changes and can be extracted as useful power. This approach is easy to manufacture using standard microfabrication processes, making it attractive for very small devices. However, it requires an

external voltage or a pre-charged material to get started, which adds complexity to the system [6, 2].

Triboelectric

Triboelectric transducers use contact between two different materials to generate surface charges. When the materials touch and then separate, charges build up and can drive a current through an external circuit. This type of harvester, known as a triboelectric nanogenerator (TENG), was first demonstrated by Fan et al.[3] and has attracted a lot of interest because it can be made from cheap, flexible materials and works well at low frequencies, such as those produced by human motion. The main weaknesses are sensitivity to humidity and wear of the contact surfaces over time [5].

Among these four mechanisms, no single technology is universally superior. The best choice depends on the specific application, the characteristics of the available vibration source, and the constraints on size and cost. This comparison motivates the investigation of alternative approaches, including variable reluctance transduction, which is discussed in the following section.

2

Rectifiers

sui [8]

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Acknowledgments

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